

Use of Virtual Reality for Emergency Service Training

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Figure 1: Excerpts from the Virtual Reality environment

ABSTRACT

This paper investigates the use of Virtual Reality (VR) in an emergency service and civil protection context. We present an exploratory user study with 30 participants sourced from the German Federal Agency for Technical Relief (THW) who were tasked with completing VR training on the assembly of a rescue platform. We evaluated the time to completion, mistakes made, cognitive load, usability, motivation, and perceived helpfulness of visual guidance elements implemented in the VR environment. The results indicate a positive effect of VR training. In addition to sourcing from a pool of emergency service workers as participants for our study, we report on the participants' feedback concerning other potential application areas for VR training in an emergency service context, which further sets our contribution apart from previous work. The paper contributes to the field of human-computer interaction by providing insights into the complex relationship between VR training and emergency service personnel.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in HCI**; **Empirical studies in visualization**; *Empirical studies in interaction design*.

KEYWORDS

Emergency Services, Civil Protection, User Study, AR/VR/XR/Immersive, Visual Guidance

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1 INTRODUCTION AND BACKGROUND

Emergency services (ES), civil defence, and disaster control forces around the world are increasingly confronted with extreme scenarios resulting from political, security policy, and ecological issues such as climate change [20]. In order to train for these scenarios, an ever wider range of training options must be available. Among a wealth of new technologies, Virtual Reality (VR) stands out as an interesting training method, particularly due to the large overlap of the application areas emergency services are facing, and the DICE framework used to determine the applicability of VR [2]. Here, VR technology may help emergency services train for scenarios that are hazardous (e.g. CBRN defence), complex (e.g. coordination of many people) or impossible (e.g. assessment of dyke damage) [15]. VR also reduces the frequency and severity of training accidents, as scenarios such as rubble landscapes or CBRN defence can be experienced immersively but do not have to be entered physically [7]. Additional benefits such as comprehensive after-action reviews, repeatable training sessions, and remote training sessions complete this array of potential applications.

VR training has shown to be a beneficial training method in a variety of environments, particularly if augmented with spatial visualisations to further enhance the training process [16, 17]. The

benefits often associated with VR training range from a reduction in time to completion (TTC) to minimising operator-induced errors [3, 16, 17, 27]. Moreover, novel frameworks, such as the Extended Reality Visual Guidance (XRVG) framework developed by Pietschmann et al. [17], may further improve this process and VR training methods overall. A limited number of previous studies have investigated the applicability of immersive technologies for use in emergency service training and operations, many of which focus on simple task execution in VR, such as fire extinguisher training [14].

These studies have shown that VR applications, regardless of whether they are used with a VR headset or on a smartphone, lead to a higher rate of knowledge acquisition, retention, and trainee confidence in performing procedures for the training of emergency personnel in safety procedures [5].

Furthermore, Lovreglio et al. [12] explore the potential benefits of integrating these technologies into the procedural training of first responders, focusing on their application in a standard course for Italian forest firefighting units. The paper concludes that integrating VR and passive haptic interfaces into training enhances engagement and effectiveness in procedural learning, especially in high-risk environments where practical skills are critical. Despite the promising advancements offered by Virtual Reality training simulations, it is essential to recognise that for optimal training effectiveness, these simulations should be complemented by hands-on, practical exercises. However, VR-based immersive training is a useful addition to this practical training and may be used to communicate or raise awareness for the risk associated with low-probability/high-impact risk (Black Swan event), like floods or flash floods [13].

Moreover, Regal et al. [18] discuss the requirements of VR training for CBRN responders. It is seen as a promising tool, but has several constraints: Firstly, the application must be deeply understood by the developers to achieve an optimal training effect. Secondly, the subject area is solved very differently, resulting in no synergy effects. Thirdly, the sweet spots for VR applications are either simple applications such as awareness training on a basic level or more complex handling procedures for specialists. Finally, there are limitations regarding haptics and interaction with the environment. While previous studies mostly focus on procedural knowledge or outlining requirements for the use of VR, we found notably fewer studies that both explored the use of VR as a training method together with ES personnel *and* investigated additional application areas beyond the one in focus for the study. Additionally, and despite VR's immense rise in presence in the academic community over the previous years, little is known about where and how it may be used to support ES training and operations. Especially concerning potential areas of application, many studies revert back to developing abstract application frameworks instead of asking the potential users – i.e. members of the emergency service community – for their expert opinion based on their years of experience in emergency response.

This study explores how VR can be used for complex training of ES personnel. Even more crucially, it puts ES workers into focus by reporting additional areas of applications as identified by ES participants after having received the training. This significantly sets our study apart from previous work. Moreover, special attention is paid to the exploration of human factors by not only measuring

TTC and mistakes but also cognitive load, usability, motivation, and the perceived helpfulness of the Visual Guidance (VG).

With this research, we hope to contribute to an improved understanding of the relationship between VR training and ES personnel and, more generally, to the field of Human-Computer Interaction in an ES context by (1) showing how VR training may be used as a beneficial tool for ES training as part of an exploratory user study with ES workers (2) inviting ES workers to identify potential areas of application for VR training based on their extensive experience (3) paving the way for future studies and developments in this field with a prototype that has kick-started the development of VR training for THW and bears the potential to be rolled out to all 80,000 ES workers from THW in Germany and beyond.

2 METHODOLOGY

To test the applicability of VR for assembly training in an ES context, a qualitative, exploratory user study was designed. It was aimed at evaluating the general applicability of VR as well as collecting user feedback relating to their views on usability, the pros and cons of these methods of training, as well as other potential application areas beyond the task selected for the initial training.

The partner of choice for the present study was the German *Federal Agency for Technical Relief* (THW), the country's federal civil protection organisation. The agency and its more than 80,000 members are involved in civil protection, humanitarian aid, as well as national and international search and rescue operations [21, 22]. The THW provided access to a pool of volunteer participants, thus offering an effective way to achieve the aforementioned goals of this exploratory user study.

2.1 Study Design

In collaboration with the agency, a suitable task first needed to be identified. Task selection was guided by several criteria: Firstly, it needed to be EM-specific but simple enough to be completed within a reasonable amount of time (approx. 10 minutes) and without any specialist knowledge that could not be conveyed as part of the training. Secondly, and based on Pietschmann et al. [16], the task was required to feature a high degree of procedural knowledge to align with the benefits offered by VR training. Lastly, the availability of assets, expert advice, and participants were taken into account.

Consequently, a rescue platform was chosen from THW's range of assembly tasks to develop a VR user study. The rescue platform consists of several scaffolding parts and requires approximately 25 steps to be assembled. It is used, for instance, to support patient extraction and rescue following lorry accidents. This selection was based on the task meeting the requirements for the study, such as representing a typical EM-like assembly task with a high degree of procedural knowledge, a medium degree of complexity, and not exceeding an estimated assembly time of 10 minutes. Additionally, the required assets (i.e. digital versions of the scaffolding parts) were available for the development of the virtual environment (VE).

Building on the XRVG framework by Pietschmann et al. [17], the VR assembly training featured two implementations of Visual Guidance elements and was aimed at obtaining participants' feedback on the visual support they received. The two implementations were targeted at supporting the picking and placing process of a

previously identified sequential assembly task: the first element represented the Object Identification dimension and was implemented as a red arrow pointing at the next part to pick up from a pool of displayed parts based on previous studies indicating this as the most obvious way to point out objects to be selected [11, 19, 24] (see Figure 2 A). The second element represented the Action Guidance dimension and consisted of a transparent hologram indicating the target location of where to add the part to the workpiece, which was also pointed out by the literature to yield positive results [10] (see Figure 2 B).

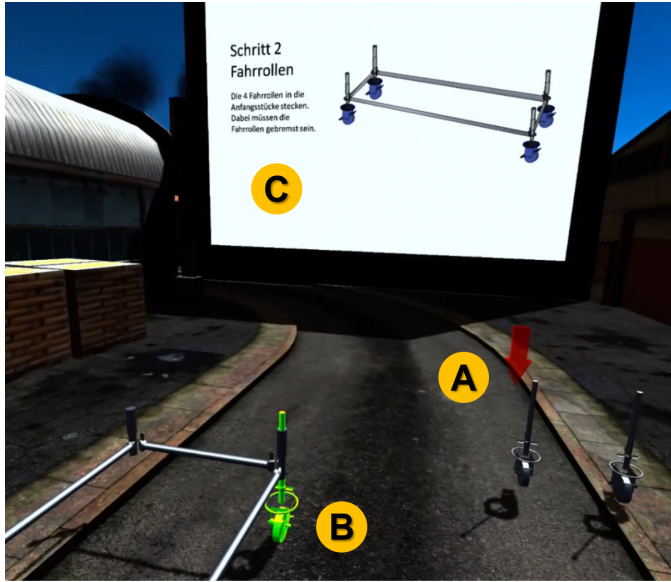


Figure 2: Virtual environment featuring the red arrow for Object Identification supporting the picking process (A), the hologram for Action Guidance supporting the placement process (B), and the instruction screen closely resembling the official (German) THW instruction manual (C) [23]

This virtual environment (for impressions see Figure 1) was developed using Unity, based on an adapted virtual environment used in a study by Farr et al. [6], and deployed on a Meta Quest 2 VR HMD. The assembly sequence followed the steps outlined in the handbook provided by the THW (see Figure 2 C). Measurements on TTC, mistakes made, CL, usability, motivation, and the perceived helpfulness of the Visual Guidance were taken to establish a benchmark for future studies. Standardised questionnaires aimed at measuring cognitive load (NASA-TLX), usability (SUS), motivation (IMI), and the perceived helpfulness of VG elements were included in the pre- and post-survey of the participants. Additionally, TTC and mistakes made were measured automatically within the VE (see subsection 2.4).

2.2 Participant Recruiting and Selection

The study was conducted in person together with the THW at two of their operation centres in Germany. Supported by representatives of the organisation, the research team visited both locations on two different weekends and asked for volunteers to participate in the

user study (opportunistic sampling). No prior knowledge of VR or a specific background (such as education or profession) was required to participate. The selection criteria required participants to:

- (1) Be over the age of 18
- (2) Have no major vision impairments (anything that would keep participants from seeing screens clearly; glasses and contact lenses are fine)
- (3) Participate only once to avoid overrepresentation of individual inputs

2.3 Study Procedure

The study itself consisted of a pre-survey, one main assembly task with 25 steps, and a post-survey. Figure 3 illustrates the study procedure, which was usually completed within 15 minutes including welcoming, briefing, and discharging participants.

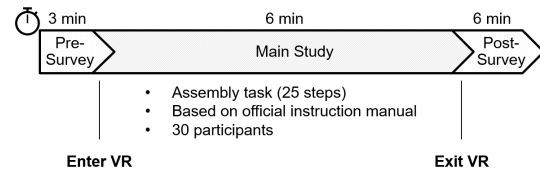


Figure 3: Procedure for the exploratory user study

Firstly, participants were briefly introduced to the study procedure and the general goals of the study, as well as given an overview of the different steps. Subsequently, participants were assigned a unique pseudonymous ID and asked to complete the pre-survey on a computer provided by the research team. Following this, they were introduced to the main study. Participants were familiarised with the controls, interaction methods, and specifics about the virtual environment before they started the timed main assembly. A large virtual monitor displayed the (assembly) instructions. These were developed based on and closely resembling THW's official instruction manual (see Figure 2 C). Lastly, upon completing the assembly task, participants progressed to the post-survey and were discharged afterwards.

2.4 Measurements

The data collected and measurements taken as part of the present study include:

- **Informed participant consent**, to confirm participants had been properly briefed and consented to participate in the study (see [1, 25, 26]).
- **Participant demographics**, such as age, occupation, and sex, which was collected as part of the pre-survey.
- **Time to completion (TTC)**, which was measured automatically by the VR HMD from the first to the last step of the main assembly.
- **Mistakes made**, which were counted automatically by the virtual environment and included the incorrect placement of correct parts and the attempted placement of incorrect parts. The mere selection of an incorrect part without an attempt to place it was not counted as a mistake.

- **Cognitive load**, measured using the NASA-TLX [8] questionnaire with a 7-point Likert scale as part of the post-survey.
- **Usability**, measured using the System Usability Scale (SUS) [4] questionnaire with a 7-point Likert scale as part of the post-survey.
- **Motivation**, measured using an adapted version of the Intrinsic Motivation Inventory (IMI) questionnaire [9] as part of both pre- and post-survey.
- **Perceived helpfulness of visual support**, which was determined using the Visual Guidance questionnaire (VisQ) to obtain insights targeted at understanding the helpfulness of the visual support).
- **Qualitative feedback**, which was collected at various points during pre- and post-surveys to give participants additional opportunity and flexibility to elaborate on their answers as well as to increase the context-sensitivity of the survey as core methodology.

All data was either collected directly within – or sent to Qualtrics from – the VE and stored in Qualtrics. The resulting dataset was exported, downloaded, and pseudonymised for local analysis.

3 FINDINGS AND ANALYSIS

A total of $n = 30$ datasets were included after checking for validity by ensuring all questions had been answered and all measurements were complete. 80% of participants were male, 20% female (0% non-confirming), and the average age was 40.9 years ($\sigma = 13.94$ years). 76.67% of participants did not report any vision issues, while 6 participants indicated minor (1 major) vision issues if not corrected through glasses. Participants were also surveyed about their educational background and occupation, as well as their experience with the scaffolding system and VR before the start of the study, all of which are displayed in Figure 4.

As their highest completed educational background, 15 (50%) indicated at least one university degree, 5 indicated A-levels, and 10 indicated a General Certificate of Secondary Education (GCSE or equivalent). Further analysis of the participants' experience with the scaffolding system used for the main assembly task and VR revealed a fairly diverse distribution across experience levels, slightly skewed towards having less experience with both.

Measurements. With the exception of mistakes made, all distributions passed the Shapiro-Wilk Test for normality. As displayed in Table 1 and visualised in Figure 5, the mean TTC was 235 seconds (just under four minutes) and the mean number of mistakes was 0.214. The median time to completion was 212 seconds (just over 3.5 minutes) and the median number of mistakes was 0.00. The mean (median) scores for cognitive load, usability, motivation, and VisQ were 28.478 (28.59), 78.935 (78.49), -0.226 (-0.333), and 4.646 (4.69) respectively.

Qualitative Feedback. Most participants expressed their satisfaction with the virtual environment and the implemented Visual Guidance, both verbally during the study as well as in the provided opportunities for free-text answers as part of the survey. The majority of participants' comments were concerned with the design of and recommendations for the training procedure, but some also included praise for the visual cues aimed at selecting the correct

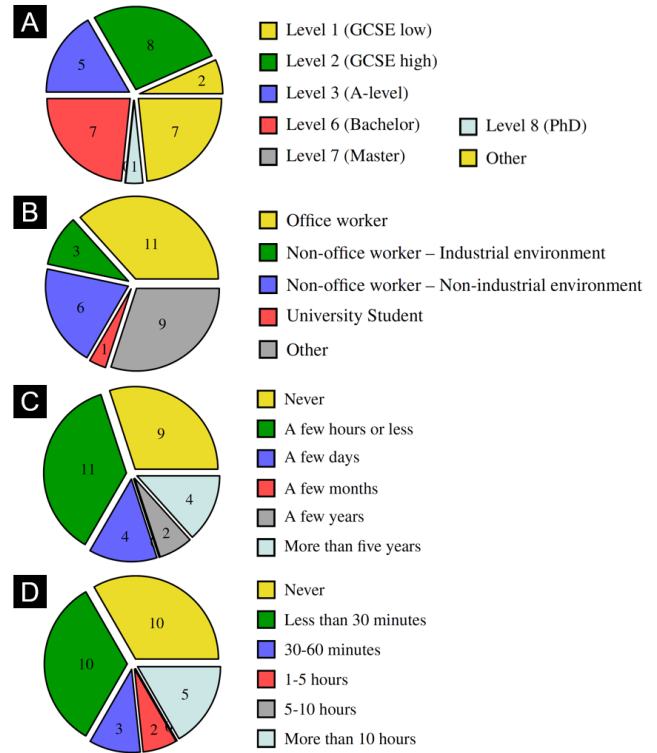


Figure 4: Participants' educational backgrounds (A), occupation (B), experience with scaffolding system (C), and experience with VR prior to the start of the study (D)

Table 1: Normality and descriptive statistics per measurement

Values	TTC	Mistakes	CL	SUS	IMI	VisQ
Normal?	Yes ($p = 0.081$)	No ($p < 0.001$)	Yes ($p = 0.092$)	Yes ($p = 0.320$)	Yes ($p = 0.276$)	Yes ($p = 0.381$)
Mean	235.33	0.214	28.478	78.935	-0.226	4.646
Median	212.0	0.0	28.58	78.49	-0.333	4.688
σ	75.909	0.418	7.275	11.718	1.112	0.799

parts, which made it easy to quickly understand the instructions and pick a part, as well as the holograms' visualisation of the target location, which made it simple to place the part (some participants even called it "impossible" to misplace a part).

Additionally, participants identified several areas as key areas for the use of VR in an ES context based on the training they had just received. Firstly, they pointed out the use for training for dangerous operations, such as handling explosives or working in close proximity to large bodies of water. Secondly, they pointed out the usefulness of VR for recreating situations and environments difficult to recreate in the real world, either because it is resource-intensive or too complex. Thirdly, the use of VR for mission briefings and de-briefings was identified. Fourthly, VR was pointed out as a useful alternative to traditional training materials such as videos to create a more immersive and capturing learning experience. Additionally, equipment familiarisation based on 3D assets, such as learning locations of equipment on their vehicle, was named

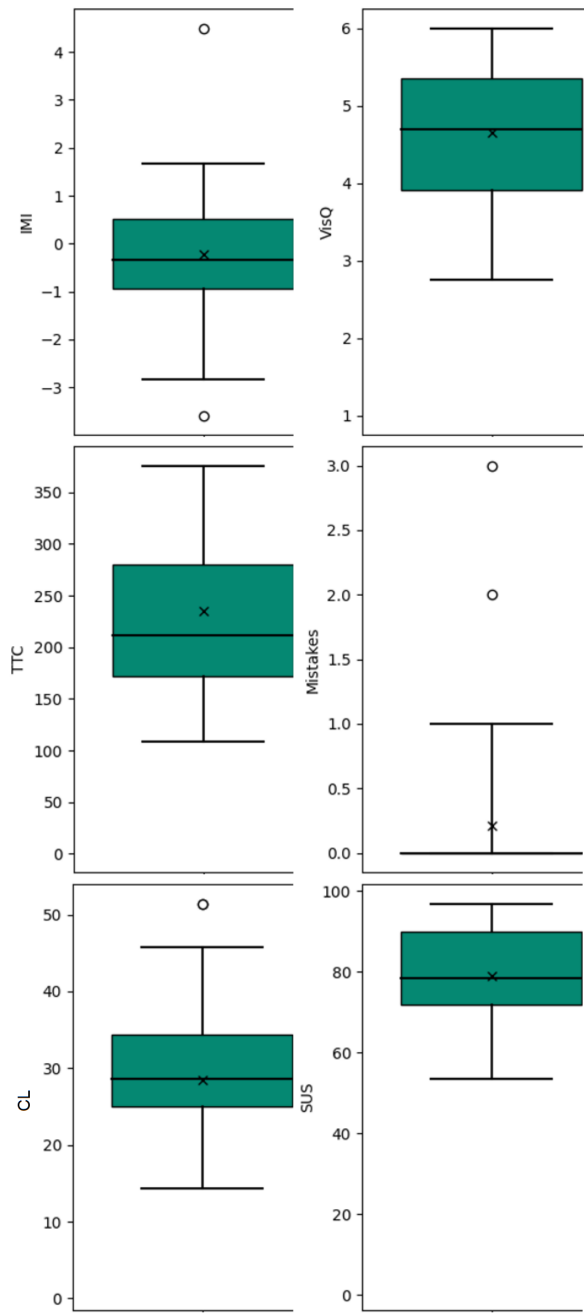


Figure 5: Boxplots for study results on time to completion (TTC), mistakes made, cognitive load (CL), usability (SUS), motivation (IMI), and perceived helpfulness of visual support (VisQ)

as a potential use case. Finally, VR was pointed out as a tool for leadership to get a more comprehensive picture by zooming out of a current mission and viewing it in bird-view mode.

4 EVALUATION AND LESSONS LEARNED

The aim of this exploratory study was twofold: firstly, we set out to investigate the usability of a VR training application with users from the emergency service community. Secondly, we gathered the users' expert opinions on other potential areas of application for VR training in an EM context more generally.

Overall, the users expressed positive feedback towards the virtual environment in general and the implemented Visual Guidance in particular. The XRVG concept supported the development and design process well and ultimately led to participants almost finding the training too easy, which is reflected in a median number of mistakes of 0.00, an overall medium cognitive load, a good usability score well above the critical mark of 70, and a high VisQ score of 4.69. The average TTC of 235 seconds was considered very low according to oral feedback received from the present instructors. Furthermore, the methodological approaches used in the present study worked well and were approved to be used again in the future. In addition to these performance metrics and human factors, the identified areas for application of VR in an ES context contain several novel contributions and pose interesting avenues for future research.

While this user study was able to achieve its aforementioned objectives, it was equally informative in highlighting potential areas of improvement and refinement in the research process. For future studies, the following 'lessons learned' have been identified based on this presented user study:

Familiarisation and Pre-Training. Including a familiarisation phase enabling participants to get used to the virtual environment and the interaction methods is useful to create fairly equal starting conditions for the pool of participants. This reduces novelty effects and bias introduced by participants' prior experience with VR (or lack thereof). While participants were given time to familiarise themselves with the VR HMD before the launch of the main application, no extra environment adhering to the pre-training principle was developed, thus requiring an individual, verbal introduction for every participant. In the future, including a separate virtual environment to standardise the familiarisation phase before the main study seems advisable to allow participants to thoroughly test the interaction method, create homogeneous starting conditions between the participants, reduce the need for repetitive explanations from the research team, and mitigate potential bias introduced by slight variations in the introduction.

Participant Diversity and Incentive. While the diverse participant pool represents one of the strengths of the present study, sourcing voluntarily (opportunistic sampling) from only two locations poses one of the limitations of this study and might have paved the way for confirmation bias to have influenced the results. Moreover, future studies should include a participant incentive to equalise motivation across the participant pool based on a common incentive as well as to simplify participant recruiting and attract participants not primed with a positive opinion towards the study, which appears to be one of the key motivators to participate from the verbal feedback received during the study. Lastly, future studies should aim to increase the share of female participants (despite the ratio achieved in this study being in line with a share of 16% female members within THW).

Complexity. Looking at a median value of 0.00 mistakes and the participants' qualitative feedback, the complexity of the assembly appears to have been too low. Consequently, it was easy for participants to avoid mistakes, resulting in potential ceiling effects. While the overall cognitive load was medium, the mistakes made are not an insightful dependent variable in its current form. Limiting the variability in scores as most participants make no mistakes reduces the meaningfulness of the results. Increasing the complexity and including more opportunities for mistakes will enable a larger spread between the participants, thus contributing to more insightful results.

Study Design. While the current study design was aimed at exploring the use of VR for emergency services as part of a qualitative user study, future study designs may want to include multiple groups to further quantify the effect of VR as a beneficial training tool as part of a between-subjects experiment. This makes observable differences between the individual groups comparable and more attributable to the different conditions, such as different VR implementations.

5 CONCLUSION

The present study investigates the use of VR for training emergency service personnel as part of an exploratory user study. The results reveal VR training to be a useful addition in an emergency service context based on measurements relating to time to completion, mistakes made, cognitive load, usability, and the perceived helpfulness of implemented visual guidance elements. The insights gained, especially as part of the free text feedback received from participants, supported the use of VR training for procedural knowledge-intensive assembly tasks and further revealed a number of additional application areas.

Our study contributes insights relating to the use of VR as a training tool for the emergency service community based on a user study with 30 emergency service members, and is among the first to investigate both the use of VR for training emergency service personnel and – crucially – potential areas of application for VR in an emergency service context coming from service members themselves after receiving the training. These findings reinforce the value VR technology may hold for supporting the important work of emergency service workers.

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